# LAPORAN JOBSHEET 2
# DATABASE OPERASIONAL

Oleh :


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**SISTEM INFORMASI BISNIS**

**POLITEKNIK NEGERI MALANG**

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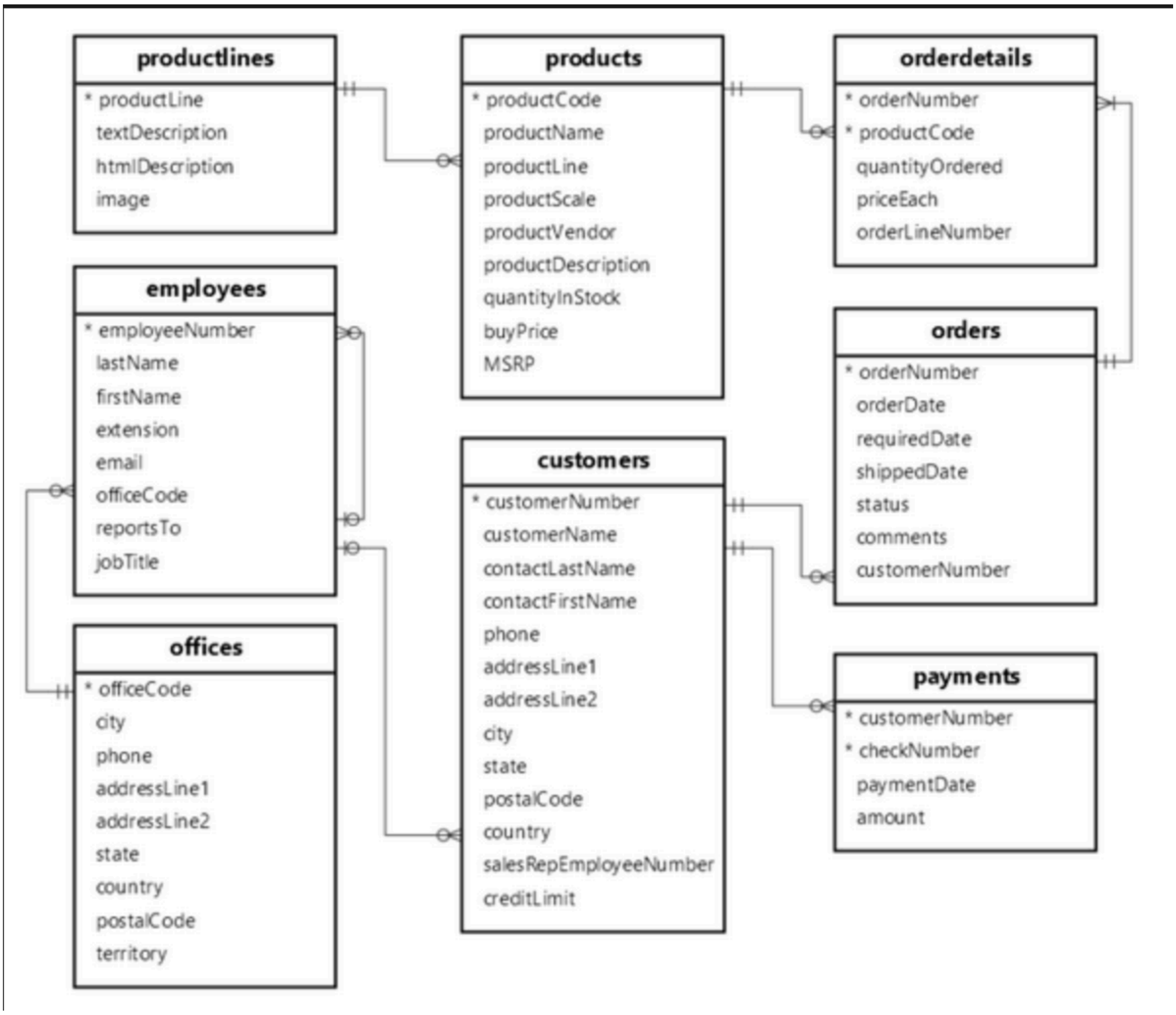
Studi Kasus

LegendVehicle merupakan perusahan jual-beli tukar-tambah kendaraan klasik. Perusahaan ini memiliki cabang di berbagai negara. LegendVehicle memiliki sistem informasi ERP sendiri. Salah satu modul dari sistem ERP tersebut adalah modul penjualan. Desain database dari modul tersebut adalah sebagai berikut:



Tugas 1

1. Import data perusahaan tersebut pada DBMS MySQL!
2. Analisa struktur data dari database perusahaan tersebut, dalam bentuk tabel, analisa hubungan setiap tabel nya!

| **Tabel 1** | **Tabel 2** | **Relasi** |
|---|---|---|
| Productlines | Products | One to Many |
| Products | Orderdetails | One to Many |
| Orderdetails | Orders | Many to One |
| Orders | Customers | Many to One |
| Customers | Payments | One to Many |
| Customers | Emplooyes | Many to One |
| Emplooyes | Office | Many to One |

3. Analisa jumlah field pada setiap tabel!

| **Nama Tabel** | **Jumlah Fileds** |
|---|---|
| Productlines | 4 |
| Products | 9 |
| Orderdetails | 5 |
| Orders | 6 |
| Customers | 10 |
| Payments | 4 |
| Emplooyes | 7 |
| Office | 9 |

A. Analisa Data

Pratikum 1

1. Jalankan query berikut pada DBMS MySql yang telah tersedia data Perusahaan LegendVehicle.

   SELECT *
   FROM employees employe, employes manager, customer cust
   WHERE employee.reportsTo=manager.employeeNumber
   AND employee.employeeNumber=cust.salesRepEmployeeNumber;

| employeeNumber | lastName | firstName | extension | email | officeCode | reportsTo | jobTitle | employeeNumber | lastName | firstName | extension | email | officeCode | reportsTo | jobTitle | customerNum |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1165 | Jennings | Leslie | x3291 | [redacted] | 1 | 1143 | Sales Rep | 1143 | Bow | Anthony | x5428 | [redacted] | 1 | 1056 | Sales Manager (NA) | |
| 1165 | Jennings | Leslie | x3291 | [redacted] | 1 | 1143 | Sales Rep | 1143 | Bow | Anthony | x5428 | [redacted] | 1 | 1056 | Sales Manager (NA) | |
| 1165 | Jennings | Leslie | x3291 | [redacted] | 1 | 1143 | Sales Rep | 1143 | Bow | Anthony | x5428 | [redacted] | 1 | 1056 | Sales Manager (NA) | |
| 1165 | Jennings | Leslie | x3291 | [redacted] | 1 | 1143 | Sales Rep | 1143 | Bow | Anthony | x5428 | [redacted] | 1 | 1056 | Sales Manager (NA) | |
| 1165 | Jennings | Leslie | x3291 | [redacted] | 1 | 1143 | Sales Rep | 1143 | Bow | Anthony | x5428 | [redacted] | 1 | 1056 | Sales Manager (NA) | |



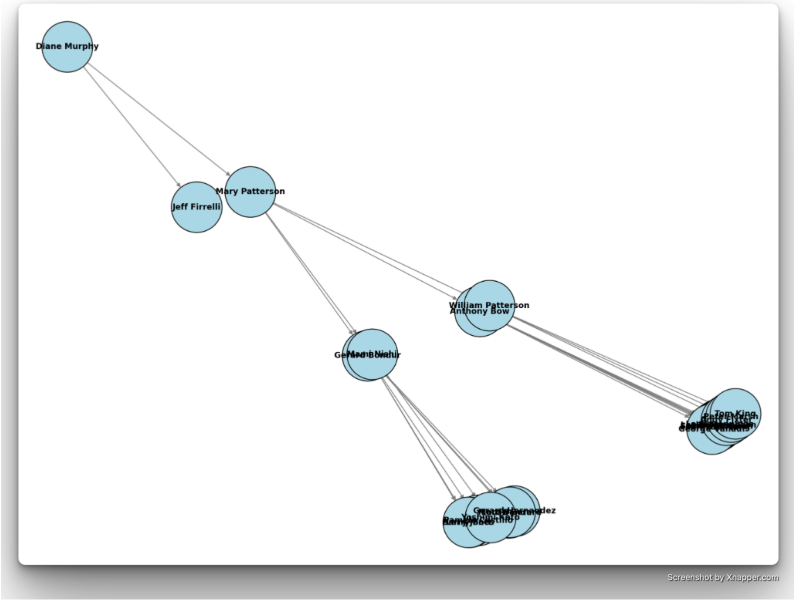
2. Buka tab baru pada browser untuk melakukan eksekusi query berikut:

   SELECT manager.employeeNumber as id_manager,
   CONCAT(manager.firstName," ,manager.lastName) as Manager,
   employee.employeeNumber as id_staff,
   CONCAT(employee.firstName," ",employee.lastName) as staff
   FROM employees employee, employees manager
   WHERE employee.reportsTo=manager.employeeNumber
   ORDER BY manager.firstName;

| id_manager | Manager | id_staff | staff |
|---|---|---|---|
| 1143 | Anthony, Bow | 1165 | Leslie Jennings |
| 1143 | Anthony, Bow | 1166 | Leslie Thompson |
| 1143 | Anthony, Bow | 1188 | Julie Firrelli |
| 1143 | Anthony, Bow | 1216 | Steve Patterson |
| 1143 | Anthony, Bow | 1286 | Foon Yue Tseng |
| 1143 | Anthony, Bow | 1323 | George Vanauf |

Tugas 2

1. Gambarlah hirarki organisasi berdasarkan atasan dari setiap pegawai sesuai dengan hasil prkatikum diatas!




2. Buka tab baru pada browser untuk melakukan eksekusi query berikut:
SELECT manager.employeeNumber as id_manager,
concat(manager.firstName," ",manager.lastName) as Manager,
employee.employeeNumber as id_staff, concat(employee.firstName,"
",employee.lastName) as staff,
count(cust.customerNumber) as total_cust
FROM employees employee join employees manager on
employee.reportsTomanager.employeeNumber
left join customers cust on
employee.employeeNumber=cust.salesRepEmployeeNumber
GROUP BY employee.employeeNumber
ORDER BY manager.firstName;

| id_manager | Manager | id_staff | staff | total_cust |
|---|---|---|---|---|
| 1143 | Anthony Bow | 1165 | Leslie Jennings | 6 |
| 1143 | Anthony Bow | 1166 | Leslie Thompson | 6 |
| 1143 | Anthony Bow | 1188 | Julie Firrelli | 6 |
| 1143 | Anthony Bow | 1216 | Steve Patterson | 6 |
| 1143 | Anthony Bow | 1286 | Foon Yue Tseng | 7 |
| 1143 | Anthony Bow | 1323 | George Vanauf | 8 |
| 1002 | Diane Murphy | 1056 | Mary Patterson | 0 |
| 1002 | Diane Murphy | 1076 | Jeff Firrelli | 0 |
| 1102 | Gerard Bondur | 1337 | Loui Bondur | 6 |

TUGAS 3

1. Siapakah staff dengan hirarki paling bawah yang berprestasi dilihat dari jumlah customer terbanyak?

```
SELECT
   employee.employeeNumber AS id_staff,
   CONCAT(employee.firstName, " ", employee.lastName) AS staff_name,
   COUNT(cust.customerNumber) AS total_cust
FROM employees AS employee
LEFT JOIN customers AS cust
   ON employee.employeeNumber = cust.salesRepEmployeeNumber
WHERE employee.employeeNumber NOT IN (SELECT DISTINCT reportsTo
FROM employees WHERE reportsTo IS NOT NULL)
GROUP BY employee.employeeNumber, employee.firstName, employee.lastName
ORDER BY total_cust DESC
LIMIT 1;
```

| id_staff | staff_name | total_cust |
|---|---|---|
| 1401 | Pamela Castillo | 10 |

Screenshot by Xnapper.com

2. Jika KPI atasan dihitung dari customer yang dimilikinya dijumlah dengan customer dari staff dibawahnya, urutkan ranking prestasi keseluruhan pegawai beserta keterangan jumlah customer yang dimilikinya!

```
WITH RECURSIVE EmployeeHierarchy AS (
   -- Ambil semua pegawai beserta jumlah customer langsung mereka
   SELECT
      e.employeeNumber AS id_staff,
      e.reportsTo AS id_manager,
      CONCAT(e.firstName, ' ', e.lastName) AS staff_name,
      COUNT(c.customerNumber) AS total_cust
   FROM employees e
   LEFT JOIN customers c
      ON e.employeeNumber = c.salesRepEmployeeNumber
   GROUP BY e.employeeNumber, e.reportsTo, e.firstName, e.lastName

   UNION ALL

   -- Gabungkan jumlah customer dengan atasan mereka
   SELECT
      m.employeeNumber AS id_staff,
      m.reportsTo AS id_manager,
      CONCAT(m.firstName, ' ', m.lastName) AS staff_name,
      eh.total_cust
   FROM EmployeeHierarchy eh
   JOIN employees m
```

```
    ON eh.id_manager = m.employeeNumber
)
-- Ambil daftar ranking KPI untuk semua pegawai
SELECT
  id_staff,
  staff_name,
  SUM(total_cust) AS kpi_total_cust
FROM EmployeeHierarchy
GROUP BY id_staff, staff_name
ORDER BY kpi_total_cust DESC;
```

| id_staff | staff_name | kpi_total_cust |
|---|---|---|
| 1002 | Diane Murphy | 100 |
| 1056 | Mary Patterson | 100 |
| 1102 | Gerard Bondur | 46 |
| 1143 | Anthony Bow | 39 |
| 1088 | William Patterson | 10 |
| 1401 | Pamela Castillo | 10 |
| 1504 | Barry Jones | 9 |
| 1323 | George Vanauf | 8 |
| 1501 | Larry Bott | 8 |
| 1286 | Foon Yue Tseng | 7 |
| 1370 | Gerard Hernandez | 7 |
| 1165 | Leslie Jennings | 6 |
| 1166 | Leslie Thompson | 6 |
| 1188 | Julie Firrelli | 6 |
| 1216 | Steve Patterson | 6 |
| 1337 | Loui Bondur | 6 |
| 1702 | Martin Gerard | 6 |
| 1611 | Andy Fixter | 5 |
| 1612 | Peter Marsh | 5 |
| 1621 | Mami Nishi | 5 |
| 1076 | Jeff Firrelli | 0 |
| 1619 | Tom King | 0 |
| 1625 | Yoshimi Kato | 0 |

3. Analisa kembali data LegendVehicle untuk mendapatkan ranking pegawai berdasarkan KPI "Jumlah omset yang didapat".

```
SELECT
    e.employeeNumber AS id_staff,
    CONCAT(e.firstName, ' ', e.lastName) AS staff_name,
    SUM(od.quantityOrdered * od.priceEach) AS total_omset
FROM employees e
JOIN customers c
    ON e.employeeNumber = c.salesRepEmployeeNumber
JOIN orders o
    ON c.customerNumber = o.customerNumber
JOIN orderdetails od
    ON o.orderNumber = od.orderNumber
GROUP BY e.employeeNumber, e.firstName, e.lastName
ORDER BY total_omset DESC;
```

| id_staff | staff_name | total_omset ▼ 1 |
|---|---|---|
| 1370 | Gerard Hernandez | 1258577.81 |
| 1165 | Leslie Jennings | 1081530.54 |
| 1401 | Pamela Castillo | 868220.55 |
| 1501 | Larry Bott | 732096.79 |
| 1504 | Barry Jones | 704853.91 |
| 1323 | George Vanauf | 669377.05 |
| 1612 | Peter Marsh | 584593.76 |
| 1337 | Loui Bondur | 569485.75 |
| 1611 | Andy Fixter | 562582.59 |
| 1216 | Steve Patterson | 505875.42 |
| 1286 | Foon Yue Tseng | 488212.67 |
| 1621 | Mami Nishi | 457110.07 |
| 1702 | Martin Gerard | 387477.47 |
| 1188 | Julie Firrelli | 386663.20 |
| 1166 | Leslie Thompson | 347533.03 |



4. Urutkan ranking pegawai beserta keterangan dana yang didapat! Jika KPI yang pertama merupakan "Jumlah customer yang bertransaksi" sedangkan KPI yang kedua "Jumlah omset yang didapat". Maka, berapakah jumlah field yang dibutuhkan untuk mendapatkan informasi tersebut?

| KPI | Jumlah Field |
|---|---|
| Jumlah customer yang bertransaksi | 3 (id_staff, staff_name, total_cust (COUNT DISTINCT customerNumber)) |

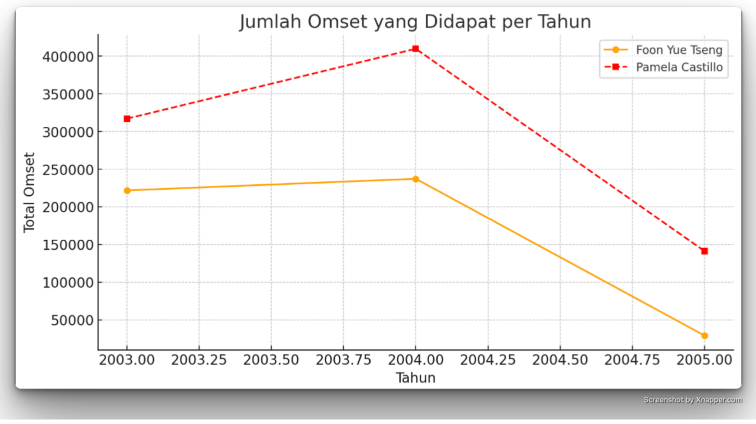
| | |
|---|---|
| Jumlah omset yang didapat | 3 (id_staff, staff_name, total_omset (SUM (quantityOrdered * priceEach)) |

5. Buatlah report pertahun untuk KPI "Jumlah omset yang didapat" pada Foon Yue Tseng dan Pamela Castillo. Serta gambarkan grafiknya (grafik garis).

```
SELECT
   e.employeeNumber AS id_staff,
   CONCAT(e.firstName, ' ', e.lastName) AS staff_name,
   YEAR(o.orderDate) AS tahun,
   SUM(od.quantityOrdered * od.priceEach) AS total_omset
FROM employees e
JOIN customers c
   ON e.employeeNumber = c.salesRepEmployeeNumber
JOIN orders o
   ON c.customerNumber = o.customerNumber
JOIN orderdetails od
   ON o.orderNumber = od.orderNumber
WHERE e.firstName IN ('Foon Yue', 'Pamela') AND e.lastName IN ('Tseng',
'Castillo')
GROUP BY e.employeeNumber, staff_name, tahun
ORDER BY staff_name, tahun;
```

| id_staff | staff_name 1 | tahun 2 | total_omset |
|---|---|---|---|
| 1286 | Foon Yue Tseng | 2003 | 221887.03 |
| 1286 | Foon Yue Tseng | 2004 | 237255.26 |
| 1286 | Foon Yue Tseng | 2005 | 29070.38 |
| 1401 | Pamela Castillo | 2003 | 317104.78 |
| 1401 | Pamela Castillo | 2004 | 409910.07 |
| 1401 | Pamela Castillo | 2005 | 141205.70 |





Studi Kasus
Pak Huhut merupakan pemegang saham LegendVehicle. dia membutuhkan dashboard untuk melihat perkembangan penjualan (omset) disetiap cabang di tiap tahunnya. Dikarenakan perusahaan tersebut belum merekrut Data Engineer maka, penarikan informasi hanya bisa dilakukan melaluai OLTP yang ada.
Analisalah terlebih dahulu:

1. Field apa saja yang diperlukan untuk menampilkan penjualan di setiap cabang
   - Cabang (Branch / Office) → Biasanya terdapat di tabel offices
     officeCode (Kode cabang)
     city (Nama kota cabang)
   - Pesanan (Orders) → Biasanya terdapat di tabel orders
     orderDate (Tanggal pesanan)
     orderNumber (ID pesanan)
   - Detail Pesanan (Order Details) → Biasanya terdapat di tabel orderdetails orderNumber (ID pesanan, untuk join dengan tabel orders)
     productCode (Kode produk)
     quantityOrdered (Jumlah produk dipesan)
     priceEach (Harga satuan)
   - Karyawan (Employees) → Biasanya terdapat di tabel employees employeeNumber (ID karyawan)
     officeCode (Kode cabang tempat karyawan bekerja)
   - Pelanggan (Customers) → Biasanya terdapat di tabel customers customerNumber (ID pelanggan)
     salesRepEmployeeNumber (ID karyawan yang menangani pelanggan)
2. Bentuk query dengan memperhatikan relasi antar tabel.

```
SELECT
  off.city AS nama_cabang,
  YEAR(ord.orderDate) AS tahun,
  SUM(od.quantityOrdered * od.priceEach) AS total_omset
FROM orders ord
JOIN orderdetails od ON ord.orderNumber = od.orderNumber
JOIN customers cust ON ord.customerNumber = cust.customerNumber
JOIN employees emp ON cust.salesRepEmployeeNumber = emp.employeeNumber
JOIN offices off ON emp.officeCode = off.officeCode
GROUP BY off.city, YEAR(ord.orderDate)
ORDER BY off.city, tahun;
```

| nama_cabang | tahun ▼ 1 | total_omset |
|---|---|---|
| San Francisco | 2005 | 378973.82 |
| Boston | 2005 | 123580.17 |
| NYC | 2005 | 101096.20 |
| Paris | 2005 | 648571.84 |
| London | 2005 | 181384.24 |
| Sydney | 2005 | 299231.22 |
| Tokyo | 2005 | 38099.22 |
| San Francisco | 2004 | 517408.62 |
| Boston | 2004 | 467177.07 |
| NYC | 2004 | 665317.99 |
| Paris | 2004 | 1465229.84 |
| London | 2004 | 706014.52 |
| Sydney | 2004 | 542996.02 |
| Tokyo | 2004 | 151761.45 |
| San Francisco | 2003 | 532681.13 |
| Boston | 2003 | 301781.38 |
| NYC | 2003 | 391175.53 |
| Paris | 2003 | 969959.90 |
| London | 2003 | 549551.94 |
| Sydney | 2003 | 304949.11 |
| Tokyo | 2003 | 267249.40 |

Screenshot by Xnapper.com

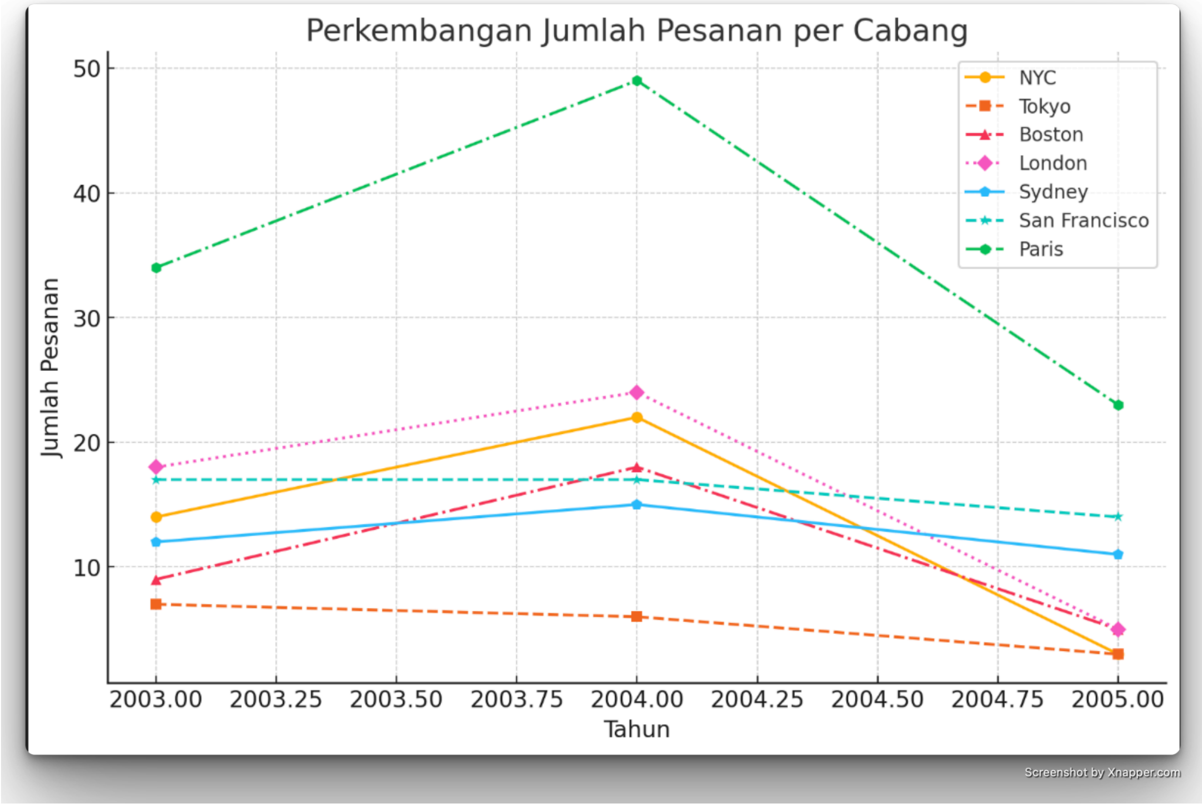
SOAL BONUS: buatlah report lain dengan sumber data OLTP yang sama, analisa field yang digunakan, bentuk struktur query dan tuliskan dalam tabel serta grafiknya.

Tabel dan Field yang Digunakan

- Tabel offices → Data cabang
  officeCode (Kode cabang)
  city (Nama cabang)
- Tabel employees → Karyawan yang bekerja di cabang tertentu
  employeeNumber (ID karyawan)
  officeCode (Cabang tempat karyawan bekerja)
- Tabel customers → Pelanggan yang ditangani karyawan
  customerNumber (ID pelanggan)
  salesRepEmployeeNumber (ID karyawan yang menangani pelanggan)
- Tabel orders → Pesanan pelanggan orderNumber (ID pesanan)
  customerNumber (Pelanggan yang melakukan pesanan)
  orderDate (Tanggal pesanan)

Struktur Query SQL

```
SELECT
   off.city AS nama_cabang,
   YEAR(ord.orderDate) AS tahun,
   COUNT(ord.orderNumber) AS jumlah_pesanan
FROM orders ord
JOIN customers cust ON ord.customerNumber = cust.customerNumber
JOIN employees emp ON cust.salesRepEmployeeNumber = emp.employeeNumber
JOIN offices off ON emp.officeCode = off.officeCode
GROUP BY off.city, YEAR(ord.orderDate)
ORDER BY YEAR(ord.orderDate), jumlah_pesanan DESC;
```


Perkembangan Jumlah Pesanan per Cabang
Jumlah Pesanan
Tahun
10
20
30
40
50
2003.00
2003.25
2003.50
2003.75
2004.00
2004.25
2004.50
2004.75
2005.00
NYC
Tokyo
Boston
London
Sydney
San Francisco
Paris
Screenshot by Xnapper.com